Unit 1: Foundations for System Development

The Systems Development Environment

System :- A system is a well-organized set of connected and working-together components that are designed to achieve a specific purpose or objective. These components can include people, processes, data, software, hardware and networks, all functioning together to turn input into useful outputs within a defined boundary. For examples, college system, Library system, etc.

Elements of the system :-

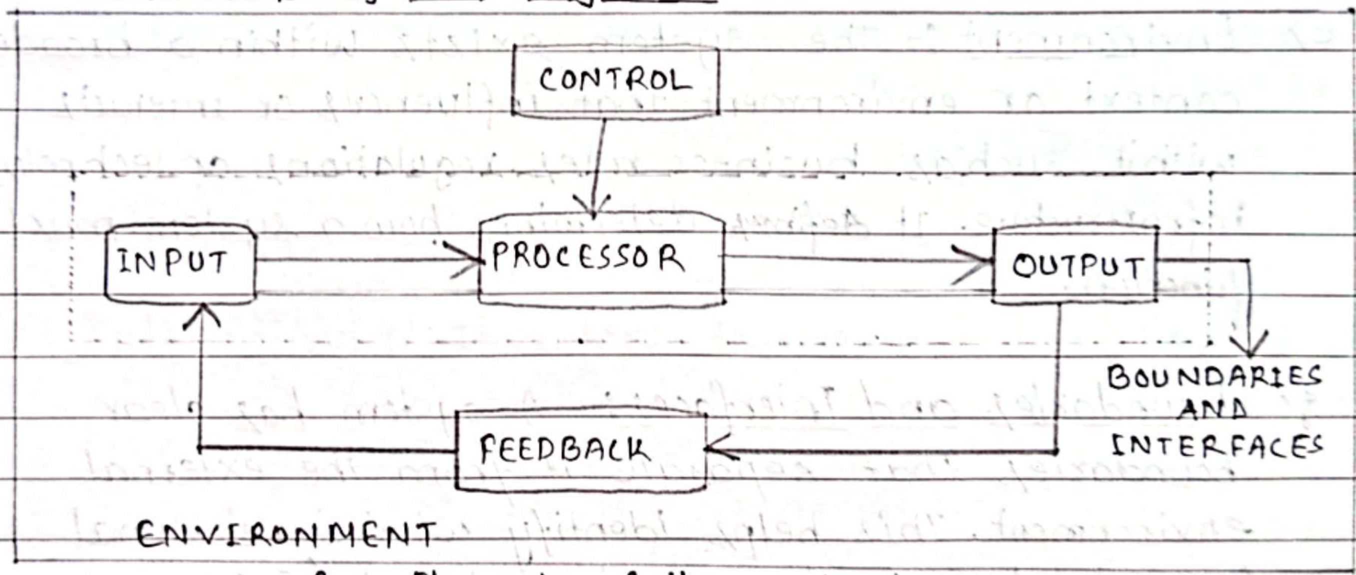


fig: Elements of the system

a) **Inputs and Outputs :-** A system takes inputs (data, commands, signals), processes them and generates outputs (information, reports, actions).

b) **Processors (Subsystem) :-** Every system is composed of several parts or sub-systems that involves the actual transformation of input into ~~oup~~ output. Processors modify the input totally or partially.

c) Control :- The control element guides the system. It is the decision-making subsystem that controls the pattern of activities governing input, processing and output.

d) Feedback :- Control in a dynamic system is achieved by feedback. Systems often include mechanism for monitoring performance and making adjustments. This is crucial for ensuring reliability and adaptability. Feedback may be positive or negative routine or informational.

e) Environment :- The system exists within a broader context or environment that influences or interacts with it, such as business rules, regulations or technological infrastructure. It ~~defines~~ determines how a system must function.

f) Boundaries and Interfaces :- A system has clear boundaries that separate it from the external environment. This helps identify what is internal (e.g. database) and what is external (e.g. users, suppliers).

Date

Characteristics of a System :-

a) Organization :- It implies structure and order. It is the arrangement of components that helps to achieve objectives.

b) Interaction :- It refers to the manner in which each component functions with other component of the system. The output of one can be the input for another.

c) Interdependence :- It means all the parts depend on each other. They are coordinated and linked together according to a plan. A change in one part affects the others and the system as a whole.

d) Integration :- It refers to the completeness of systems. It is concerned with how a system is tied together. The components work together as a whole rather than independently. The system performs better than the sum of individual parts.

e) Central Objective :- Every system has a purpose or goal that guides its functioning and existence. Objectives may be real or stated. Although a stated objective may be the real objective.

Information System :- An Information System is an integrated set of components - including hardware, software, data, processes and people - that work together to collect, process, store and distribute information to support decision-making, coordination, control, analysis and visualization within an organization.

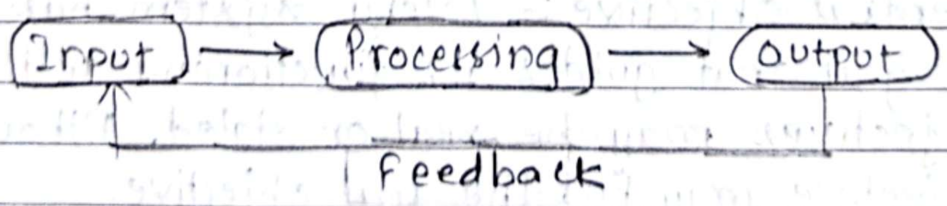
➤ Information system capture and manage data to produce useful information that supports an organization and its employees. Data are the collection of raw facts representing events occuring in organizations. Information is the data shaped into a form that is meaningful and useful.

Three activities to prodoce information in an information system are input, processing and output.

Input captures or collects raw data from within the organization or from its external environment.

Processing converts, manipulates, and analyze these raw data into the meaningful information.

Output transfers this information to the people who will use it or to the activities for which it will be used.



feedback is output that is returned to the appropriate members of the organization to helpthem evaluate or correct input.

Date

Typical Components of Information Systems :-

1. **People** :
   - → Users who interact with the system
   - → IT professionals who build and maintain it.
   - → Decision-makers who depend on it.

2. **Processes (procedures)** : Defined workflows and procedures that guide how data is collected, managed and transformed into meaningful Output.

3. **Technology** :
   - → Hardware (e.g., servers, computers, networking devices)
   - → Software (e.g., database systems, enterprise applications).

4. **Data** :
   - → Raw facts that are processed into useful information.
   - → Structured (like in databases) or unstructured (like emails or documents)

5. **Networks** : Enable communication and data sharing across different parts of the system and organization.

Date

Types of Information System :

1> Transaction Processing Systems :- Transaction processing systems (TPS) are computerized inform--ation systems that were developed to process large amounts of data for routine business transactions such as payroll and inventory. A TPS eliminates the tedium of necessary operational transactions and reduces the time once required to perform them manually, although people must still input data to computerized systems

2> Office Automation Systems and Knowledge Work Systems
At the knowledge level of the organization are two classes of systems.

office automation systems (OAS) support data workers who do not usually create new knowledge but rather analyze information to transform data or manipulate it in some way before sharing it with or formally disseminating it throughout, the organization and sometime beyond. Some fundamental aspects of OAS are word processing, spreadsheets, desktop publishing, electronic scheduling etc.

Knowledge Work Systems (KWS) support profession--al workers such as scientists, engineers and doctors by aiding them to create and share new knowledge.

3> Management Information Systems (MIS) :- A Management Information System (MIS) is an integrated system that gathers, processes, stores and disseminates information to support informed decision making within an organization. It uses technology and people to analyze data, identify trends, and provide insights

Date

that help managers at all levels make better decisions. MIS is used for decision-making and for the coordination, control, analysis and visualization of information in an organization.

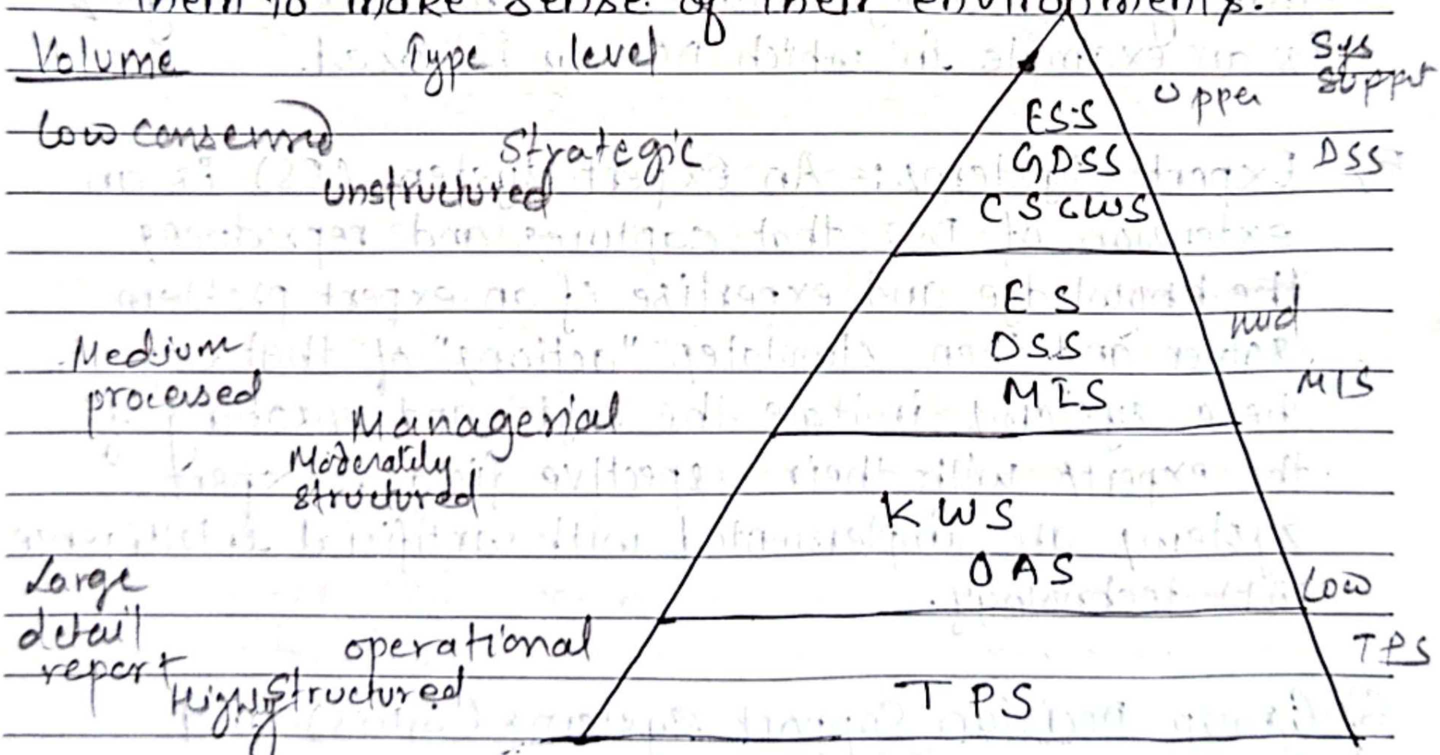
4) Decision Support Systems :- These systems also serve at the management level of the organization. These systems use internal information from TPS and MIS, and often information from external sources. DSS have more analytical power than other systems. DSS support complex decision-making with analysis tools. Contract cost analysis is an example in which DSS can be used.

5) Expert Systems:- An Expert system (ES) is an extension of DSS that captures and reproduces the knowledge and expertise of an expert problem solver and then simulates "actions" of that expert. These systems imitate the logic and resoning of the experts with their respective fields. Expert systems are implemented with artificial intelligence (AI) technology.

6) Group Decision Support Systems (GDSS) and Computer-Supported Collaborative Work Systems (CSCWS) :- GDSS is an information system that is intended to bring a group together to solve a problem with the help of various supports such as polling, questionnaires, brainstorming, and scenario creation.

Sometimes GDSS are discussed under more general term Computer-Supported Collaborative Work Systems (CSCWS), which might include software support called groupware for team collaboration via networked computers.

7) Executive Support Systems :- These systems serve the strategic level of the organization. ESS helps executives organize their interaction with the external environment. Although ESS rely on the information generated by TPS and MIS, Executive support systems help their users address unstructured decision problems, which are not application specific. ESS extends and supports the capabilities of executives, permitting them to make sense of their environments.



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System Analysis and Design :- System Analysis and Design (SAD) is a structured process used to develop information systems that effectively support business processes and user needs. It encompases the study of existing systems, the identification of requirements and the logical design and implementation of improved systems.

System Analysis :- This is the investigative phase where the focus is on understanding what the system must do.

System Design :- This phase focus on how the system will meet the identified requirements.

In systems analysis and design, we use various methodologies, techniques and tools.

Methodologies :- Comprehensive, multi-step approaches to systems development.

Techniques :- Processes that are followed to ensure that work is well thought -out, complete and comprehensibe to others on the project team.

Tools :- Computer programs to assist in application of techniques to the analysis and design process.

Importance

→ Improves system quality
→ Enhances user satisfaction
→ Supports informed decision making
→ Mitigates risk.

System Analyst :- A system analyst is a senior most person in the system development hierarchy, who analyses the system and takes the responsibility for development of computer based new information system for the end users. The system analyst is an agent of change and innovation having the primary responsibility of system analysis and design.

Job Responsibilities :-

→ Implements system requirements by defining and analyzing system problems; designing and testing standards and solutions.

→ Defines application problem by conferring with clients, evaluating procedures and processes.

→ Develops solution by preparing and evaluating alternative workflow solutions.

→ Controls solution by establishing specifications and coordinating production with programmers.

→ Validates results by testing programs.

→ Ensures operation by training client personnel and providing support.

→ Provides reference by writing documentation.

→ Accomplishes information systems and organization mission by completing related results as needed.

Skills (characteristics)

a) Communication
b) Farsightedness and vision
c) Adaptability
d) flexibility
e) Patience and rationality
f) Sound temperament
g) Management skills
(h) Creativity & Problem Solving
(i) Technical knowledge
(j) strong analytical skill
(k) Training and documentation capability.

Date

A Modern Approach to System Analysis and Design:

The growth of computer-based information systems analysis and design methodologies started during the year 1950 to 1960. The researchers argued that software crisis was due to the lack of discipline of programmers and some believed that if formal engineering methodologies would be applied to software development, then production of software would become as predictable.

The new technologies and practices which were developed after 1970-1990 were primarily focused on solving the software issues like software crisis. The major elements used were software tools, formal methods, well defined processes that use the methodologies like OOP, CASE tools and structured programming approaches.

DEVELOPING INFORMATION SYSTEM AND THE SYSTEM DEVELOPMENT LIFE CYCLE.

The system development life cycle is classically thought as the set of activities that analysts, designers and users carry out to develop and implement an information system. It creates the framework carefully structuring, planning and controlling the process of developing an information system which fulfills the demands of information needs. i.e. The systematic approach of developing any computer based system comprising different phases is called system development life cycle. SDLC facilitates standardization and management control of the development process. SDLC typically comprises following phases:-

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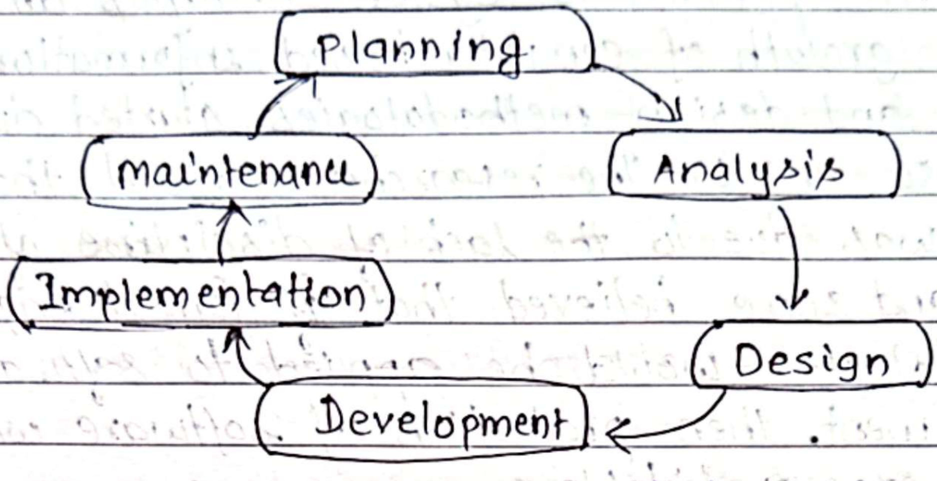


1) Planning :- The first phase is called planning. In this phase, an organization's total information system needs are identified, analyzed, prioritized, and arranged.

2) Analysis :- This is the most important phase in SDLC in which the system analyst throughly studies current system and all the information is gathered from customers, users and other stake-holders. This phase has two sub-phases.

i) Requirement Determination :- In this sub-phase analyst is involved in careful study of current system to find the problem and limitation of the current system, communicate with different stakeholders and identifies the requirements for the new system.

ii) Analysis :- In this part the system analyst study the requirements and the best alternative solution for the fullfillment of user's requirement is selected. The output of ~~all~~ the analysis phase is a description of alternative solution. In this phase analyst performs Need analysis and feasibility analysis.

3) <u>Design</u>: This is the most creative and challenging phase of SDLC. In this phase the analyst convert description of the recommended alternative solution into logical then physical system specification.
To develop the logical design of system, functional specification of system and details about the computer configuration is taken as input. During this phase the logic of programs, files or databases are designed, program test plans and implementation plans are drawn up.

4) <u>Development</u>: In this phase the designs of the systems are actually implement using programming languages, programmer writes the code for the components or may use CASE tools for building the components.
The individual components are then tested for errors and defects and are fixed. Once all components are developed and tested individually they are integrated and test Integration testing is performed followed by System testing and User acceptance testing.

5) <u>Implementation</u>:- After development and testing of the system, implementation takes place. This phase includes the installation of the hardware before the system is handed over to the user. It covers

→ User training → Data Conversion
→ Control procedure for the change over.

Date

6) Maintenance :- This is the longest phase in SDLC. In this phase information system is systematically repaired and improved. Once the system is deployed and customers start using the developed system, following 3 activities occur:

→ Bug fixing → Upgrade → Enhancement

The main focus of this phase is to ensure the needs continue to be met and that the system continues to perform as per the specification mentioned in the Analysis phase.

The Heart of the System Development Process:

In traditional SDLC practice analysis, design and implementation ~~were~~ are considered distinct phases. Current practice combines these activities. Insted of systems requirements being produced in analysis, systems specifications being created in design, and coding and testing being done at the beginning of implementation, current practice combines all of these activities into a single analysis-design-code-test process. These activities are the heart of systems development. This combination of activities is typical of current practices in Agile Methodologies.

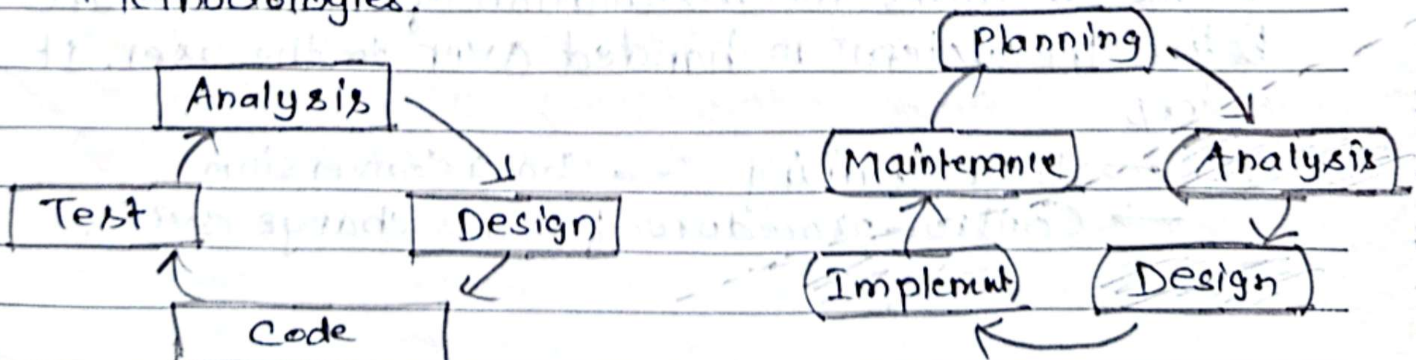


fig:- The analysis-design-code-test loop

fig :- The heart of System develop-

Date

Traditional Waterfall SDLC

Waterfall Model is the simplest, and first to be offered as a Process Model. In a Waterfall model, each phase must be finished before the next one can begin and the stages must not overlap. The entire system development is separated into phases in this technique, at the completion of each phase, a milestone has been reached and a document is produced to be approved by the stakeholders before moving to the next phase, painstaking amounts of documentation and signoffs through each part of the development cycle is required.

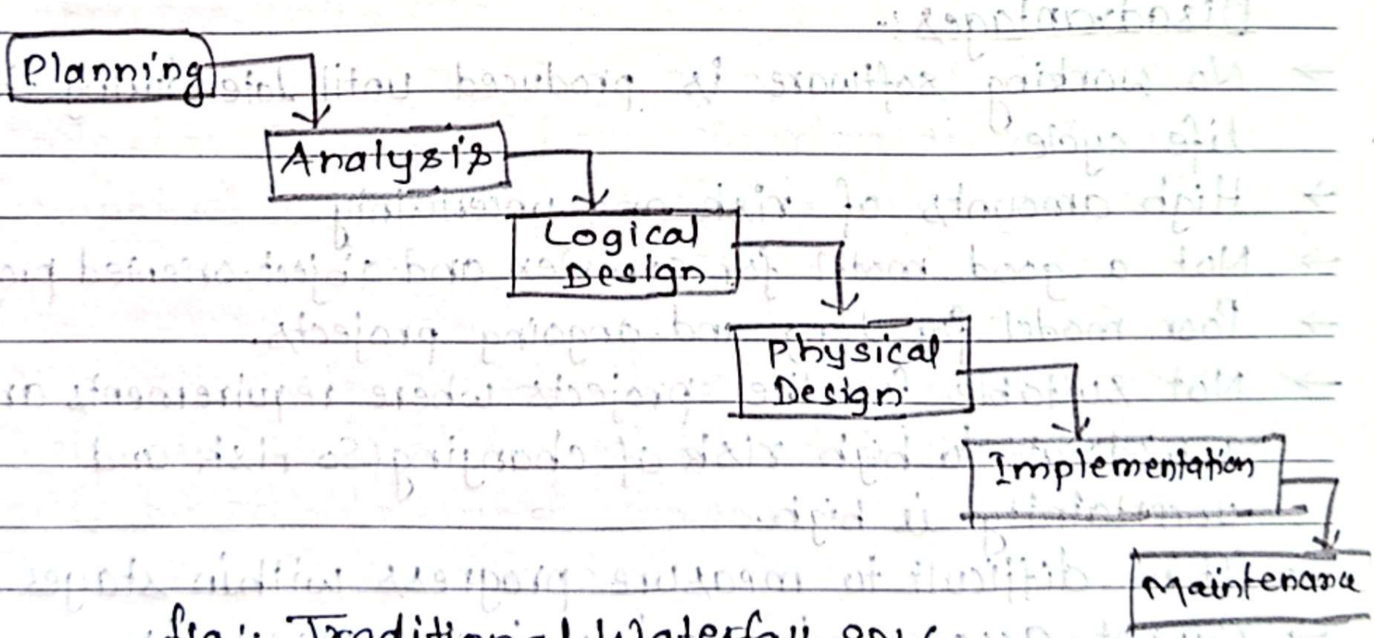


fig:: Traditional Waterfall SDLC

Some Situations where the use of Waterfall Model is most appropriate :-

→ Requirements are very well documented, clear and fixed.
→ Product definition is stable.
→ Technology is understood and is not dynamic.
→ There are no ambiguous requirements.
→ Ample resources with required expertise are available to support the product.
→ The project is short.

Advantages of Waterfall Model:

→ Simple and easy to understand and use.
→ Easy to manage due to the rigidity of the model.
  – each phase has specific deliverables and a review process.
→ Phases are processed and completed one at a time.
→ Works well for smaller projects where requirements are very well understood.
→ clearly defined stages.
→ Well understood milestones.
→ Easy to arrange tasks.
→ Process and results are well documented.

Disadvantages:-

→ No working software is produced until late during the life cycle.
→ High amounts of risk and uncertainty.
→ Not a good model for complex and object-oriented projects.
→ Poor model for long and ongoing projects.
→ Not suitable for the projects where requirements are at a moderate to high risk of changing. So risk and uncertainity is high.
→ It is difficult to measure progress within stages.
→ Cannot accommodate changing requirements.
→ ~~No~~ Adjusting scope during the life cycle can end a project.
→ Integration is done as a "big-bang" at the very end, which doesn't allow identifying and technological or business bottleneck or challenges early.
→ Costly and requires more time in addition to the detailed plan.

Date

CASE Tools

Computer-Aided Systems Engineering (CASE), also called computer-aided software engineering, is a technique that uses powerful software, called CASE tool, to help systems analysts develop and maintain information systems. CASE tools provide an overall framework for systems development and support a wide variety of design methodologies, including structured analysis and object oriented analysis. CASE tools can be used to help in the project identification and selection, project initiation and planning, analysis, and design phases and/or in the implementation and maintenance phases of SDLC.

Types of CASE tools :-

→ Diagram Tools : It helps in diagrammatic and graphical representations of the data and system processes.

→ Computer Display and Report Generators :- It helps in understanding the data requirements and the relationships involved.

→ Analysis Tools :- It automatically checks for incomplete, inconsistent, or incorrect specifications in diagrams, forms, and reports.

→ Central Repository : It provides the single point of storage for data diagrams, reports and documents related to project management.

→ Documentation Generators :- It helps in generating technical and user documents in standard format.

→ Code Generators :- Code generators enable the automatic generation of program and database definition code directly from the design documents, diagrams, forms and reports.

* Advantages :
→ As special emphasis is placed on redesign as well as testing, the servicing cost of a product over its expected lifetime is considerably reduced.

→ The overall quality of the product is improved as an organized approach is undertaken during the process of development.

→ Chances to meet real-world requirements are more likely and easier with a computer-aided software engineering approach.

→ CASE indirectly provides an organization with a competitive advantage by helping ensure the development of high-quality products.

Disadvantages

→ Expensive (high cost).

→ Learning Curve. (slow initial phase)

→ Tool Mix. (Choosing appropriate tool)

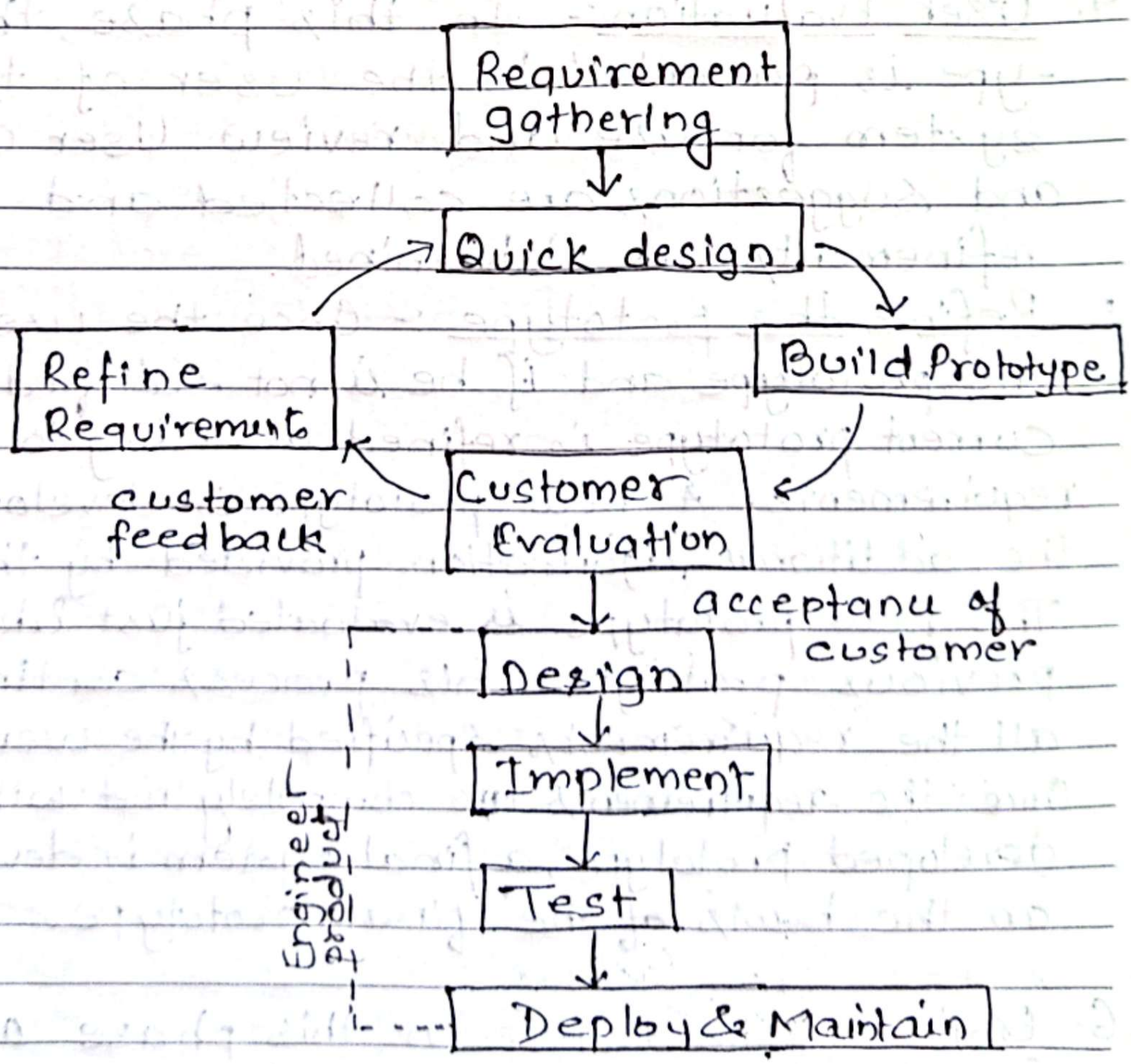
Date

Other Approaches

Prototyping Approach:-

A prototype model is a preliminary sample, model, or release of a product used to test an idea or method.

This type of System Development method is employed when it is very difficult to obtain exact requirements from the customer. While making the model, users keeps giving feed-backs from time to time and based on it, a prototype is made. After completion of this, a more accurate system requirement specification (SRS) is prepared.



Phases of Prototyping Model :-

1. Requirements Gathering :- This is the first phase of prototyping model. In this, the system's service, constraints and goals are established by consultation with system users.
2. Quick design: On the basis of known requirement, a preliminary design or quick design for the system is created.
3. Build Prototype :- In this phase, the information from the design is rolled into prototype, which represents the working model of the required system.
4. User Evaluation:- In this phase, the prototype is presented to the user of the system for use and review. User comments and suggestions are collected and areas of refinements are determined.
5. Refine the prototype :- Once the user evaluates the prototype and if he is not satisfied, the current prototype is refined according to the requirements. A new prototype is developed with the additional information provided by the User. The new prototype is evaluated just like the previous prototype. This process continues until all the requirements specified by the user are met. Once the requirements are completely met with the developed prototype, a final system is developed on the basis of the final prototype.

6. Engineer Product :- In this phase a final design is prepared on the basis of the accepted prototype, which is then realized and tested

Date

as a new system. The final system is evaluated throughly followed by the routine maintenance on regular basis.

Advantages:

→ The customers get to see the partial product early in the life cycle. This ensures a greater level of customer satisfaction and comfort.

→ New requirements can be easily accommodated as there is scope for refinement.

→ Missing functionalities can be easily figured out.

→ Errors can be detected much earlier, saving a lot of effort and cost, besides enhancing the quality of the Software.

→ The developed prototype can be reused by the developer for more complicated projects in the future.

→ flexibility in design.

Disadvantages:

→ There are no parallel deliverables.

→ It is a time consuming if customers ~~ak~~ asks for changes in prototypes.

→ This methodology may increase the system complexity as scope of the system may expand beyond original plans.

→ The invested effort in the preparation of prototypes may be too much if not properly monitored.

→ Customer may get confused in the prototypes and real systems.

Date

Spiral Approach: The spiral development model is a risk-driven processes model generator that is used to guide multi-stakeholder concurrent engineering of software intensive systems. It has two main features: One is cyclic approach for incrementally growing a system's degree of definition and implementation while decreasing its degree of risk. The other is set of anchor point milestones for ensuring stakeholder commitment to feasible and mutually satisfactory system solutions.

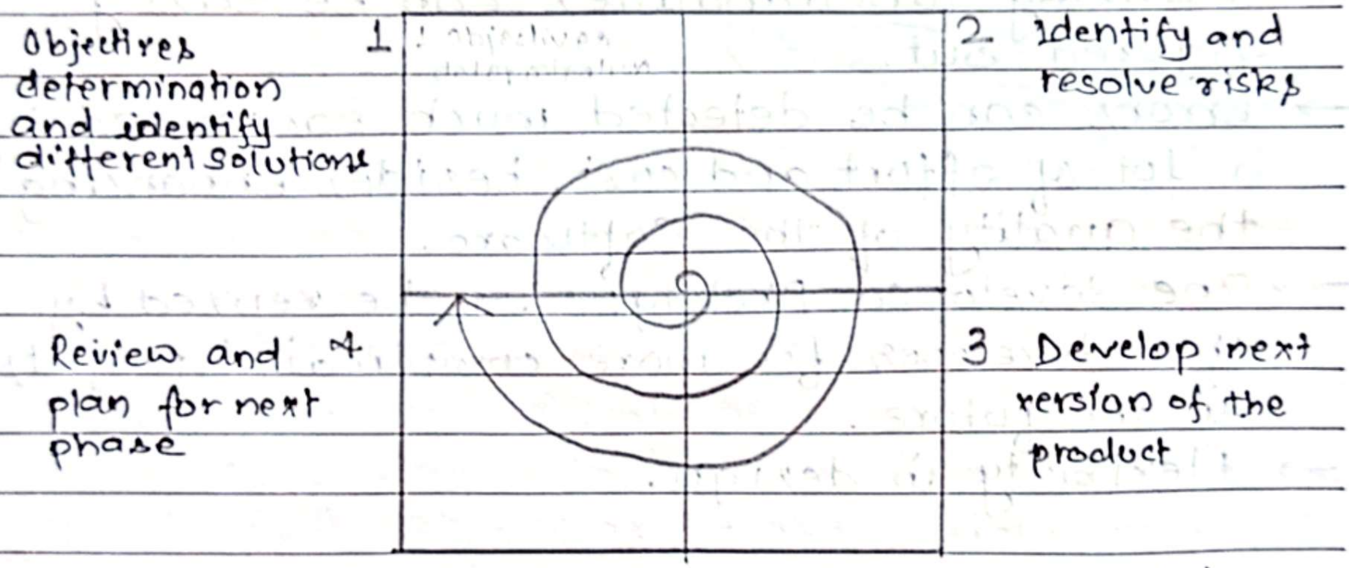


fig:- Different phases of spiral model.

1. Objectives determination and identify alternative solutions: Requirements are gathered from the customers and the objectives are identified and analyzed. Then alternative solutions possible for the phase are proposed in this quadrant.

2. Identify and resolve risks:- The risks associated with the solution are identified and the risks are resolved using the best possible strategy.

Date

3. Develop next version of the product: During this third quadrant, the identified features are developed and verified through testing. At the end of the third quadrant, the next version of the software is available.

4. Review and plan for the next phase: In the fourth quadrant, the customers evaluate the so far developed version of the software. In the end, planning for the next phase is started.

Advantages:

→ High amount of risk analysis hence, avoidance of Risk is enhanced.
→ Good for large and mission-critical projects.
→ Strong approval and documentation control.
→ Additional Functionality can be added at a later date.
→ Software is produced early in the software life cycle.

Disadvantages:-

→ Can be a costly model to use.
→ Risk analysis requires highly specific expertise.
→ Project's success is highly dependent on the risk analysis phase.
→ Doesn't work well for smaller projects.

Rapid Application Development Approach (RAD)

Rapid application development (RAD) is an object-oriented approach to systems development that includes a method of development as well as software tools. RAD is trying to meet rapidly changing business requirements more closely. To deliver an application to the Web before their competitors, businesses may want their

Date

development team to experiment with RAD.

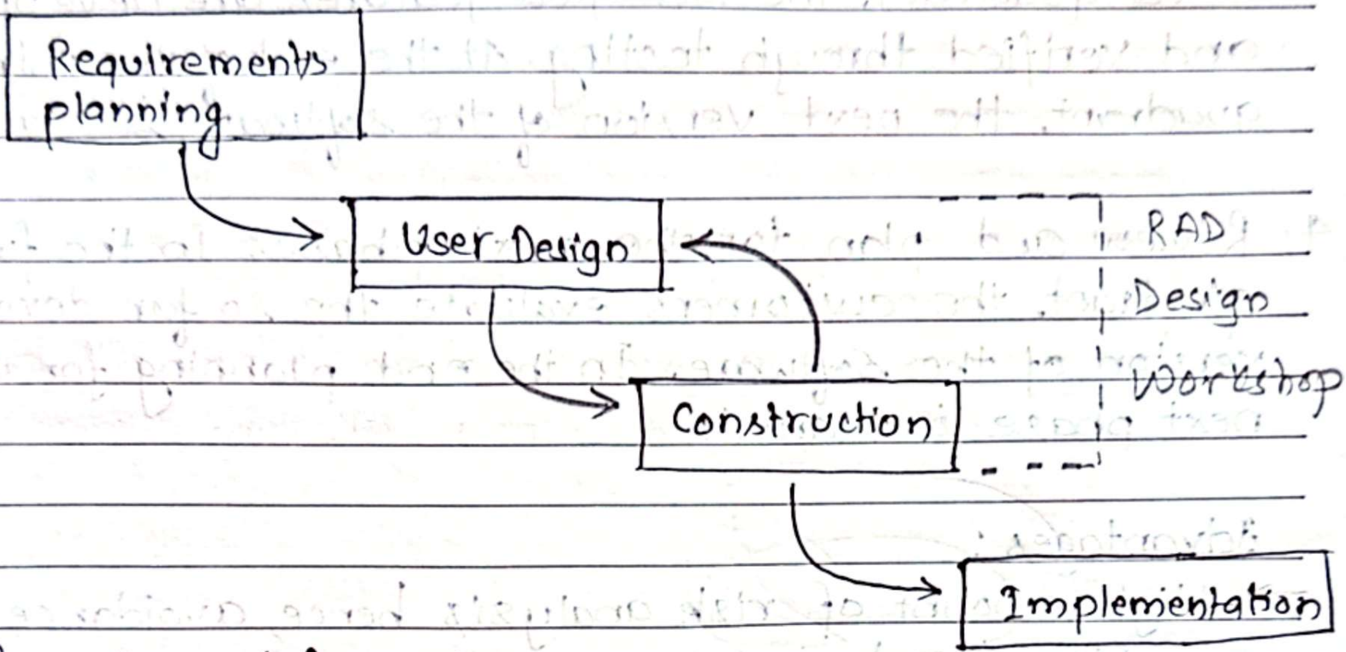


fig: RAD life Cycle

phases:

i) Requirements Planning :- In this phase, users and analysts meet to identify objectives of the application or system. It may involve users from different levels of the organization. We may be working with the chief Information officer (CIO) as well as strategic planners.

ii) RAD Design Workshop :- The RAD design workshop phase is a design and refine phase that can best be characterized as a workshop. During RAD design workshop, users respond to actual working prototypes and analysts refine designed modules based on user responses.

iii) Implementation :- Analysts work with users to design the business or non-technical aspects of the system. As soon as these aspects are agreed on and the systems are

Date

built and refined.

Advantages :-

→ changing requirements can be accommodated.
→ Progress can be measured.
→ Iteration time can be sort with use of powerful RAD tools.
→ Productivity with fewer people in short time.
→ Reduce development time.
→ Increases reusability of components.

Disadvantages :-

→ Dependency on technically strong team members for identifying business requirements.
→ Only system that can be modularized can be built using RAD.
→ Requires highly skilled developers/designers.
→ High dependency on modeling skills.
→ Inapplicable to cheaper projects as cost.

Agile Development Approach

Agile software development is a system development model based on iterative and incremental development in which requirement and solutions evolve through collaboration between self organizing, cross functional teams. Agile is a term used to describe system development approaches that employ continual planning, learning, improvement team collaboration, evolutionary development and early delivery.

Date

Manifesto for Agile Model:-

→ Individuals and interactions over processes and tools.
→ Working software over comprehensive documentation.
→ Customer collaboration over contract negotiation.
→ Responding to change over following a plan.

~~Ag~~ In agile model items in left are valued more than the items in right.

Principles:-
i) Customer Involvment
ii) Incremental delivery
iii) People not process
iv) Embrace change
v) Maintain simplicity.

fig:- Sprint outcome

Agile Methodologies ~~is~~ are not for every project. Agile or adaptive process is recomended if project involves:
→ Unpredictable or dynamic requirements.
→ Responsible and motivated developers.
→ Customers who understand the process and will get involved

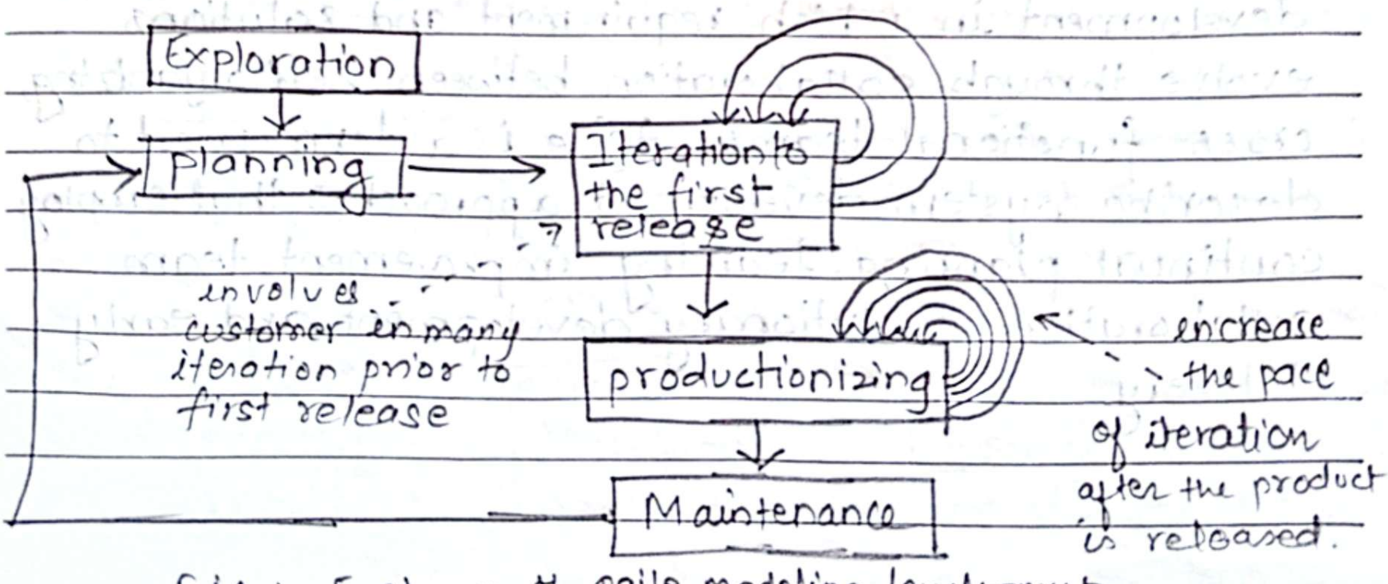


Fig:- 5 stages of agile modeling development

Date

Phases:-

1) Exploration:- During this we practice estimating the time needed for variety of tasks. This is all about the curious attitude toward the work environment, its problems, technologies and people.

2) Planning: In this phase we and customers agree on a date anywhere from two months to half a year from the current date to deliver solutions to their business problems.

3) Iterations to the first release:- In this phase we will sketch out the entire architecture of the system, even though it is just outline or skeletal form.

4) Productionizing: The product is released in this phase, but may be improved by adding other features.

5) Maintenance: Once the system has released, it needs to be kept running smoothly.

Advantages:

→ Working through pair programming produce well written compact programs which has fewer errors as compared to programmers working alone.
→ Customer satisfaction by rapid, continuous delivery of useful software.
→ It provides best communication among customers, developers and testers.
→ Continuous attention to technical excellence and good design.
→ Regular adaptation to changing circumstances.
→ Even late changes in requirements are welcomed and will not affect to develop a system.

Disadvantages

→ It takes more time and energy for everyone involved.

→ It has lack of necessary documentation.

→ The project can easily get taken off track if the customer representative is not clear what final outcome that they want.

→ Due to absence of proper documentation, when the project completes and the developers are assigned to another project, maintenance of the developed project can become a problem.

Service-Oriented Architecture (SOA)

One of the newer current approaches to systems development is called Service-Oriented Architecture (SOA). The idea behind SOA is to build systems around generic services, or specific business functions, which can be used in many different applications. Once a set of services have been identified and validated, developers can assemble them into new application. Developing a service is typically done by taking standard object access protocol.

An example of a service is a credit check. Any application that deals with customers or suppliers would have a credit check function. In an SOA approach, one of the existing credit check functions would be converted into a service that all current and future application could use. There would be no need to model the same function yet again.

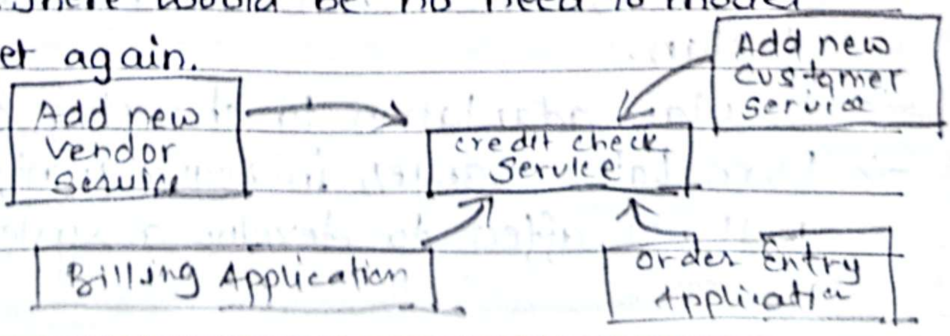


fig:- Illustration of a service, a credit check, used by application and other services

Date

eXtreme Programming:

eXtreme Programming is an agile approach to software development which is distinguished by its short cycles, incremental planning approach, focus on automated tests written by programmers and customers to monitor the development process, and a reliance on an evolutionary approach to development that lasts throughout the lifetime of the system. Key emphases of extreme Programming are its use of two-person programming teams and having a customer on-site during the development process.

In this approach planning, analysis, design and construction are all fused into a single phase of activity. Under this approach, coding and testing are intimately related parts of the same process. Code is tested very soon after it is written. Part of eXtreme programming that makes the code-and-test process work more smoothly is the practice of pair programming. All coding and testing is done by two people working together who write code and develop tests.

Advantages of pair programming are

1. more (and better communication among developers.
2. Higher levels of productivity.
3. Higher quality code
4. Reinforcement of the other practices in eXtreme programming

Although the extreme programming process has its advantages, just as with any other approach to systems development, it is not for every one and is not applicable to every project. They are mainly applied for

(i) Small projects.
(ii) Projects involving new technology or Research projects.

Date

Object Oriented Development (OOD)

Object-Oriented Analysis and Design

OOAD is often called the third approach to systems development, after the process oriented and data-oriented approaches. The object-oriented approach combines data and processes (called methods) into single entities called objects. Objects usually correspond to the real things an information system deals with such as customers, suppliers, contracts etc. The goal of OOAD is to make systems elements more reusable, thus improving system quality and the productivity of systems analysis and design. Another key idea behind object orientation is inheritance. Objects are organized into object classes, which are groups of objects sharing structural and behavioral characteristics.

Date

Managing the Information Systems Project

Project management is an important aspect of the development of information systems and a critical skill for a systems analyst. The focus of project management is to ensure that systems development projects meet customer expectations and are delivered within budget and time constraints.

Shaping a project

A successful project must be completed on time, within budget, and deliver a quality product that satisfies users and meets requirements. Project management techniques can be used throughout the SDLC. Systems development projects tend to be dynamic and challenging. There is always a balance between constraints, and interactive elements such as project cost, scope, and time.

Project Triangle

For each project, it must be decided what is important, because the work cannot be good and fast and cheap. When it comes to project management, things are not quite so simple. The challenge is to find the optimal balance among the three factors cost, scope and time. Any change in one of the factor will affect other two. Most successful Project Managers rely on personal experience, communication ability and resourcefulness.

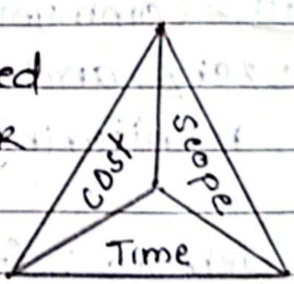


fig:- A typical Project triangle including cost, scope & time.

Date

Project Manager

The project manager is a systems analyst with a diverse set of skills - management, leadership, technical, conflict management and customer relationship - who is responsible for initiating, planning, executing, and closing down a project. Creating and implementing successful projects requires managing the resources, activities, and tasks needed to complete the information systems project for that project manager requires various sets of skills.

Common Activities and skills of a project manager:

a) Leadership : Influencing the activities of others toward the attainment of a common goal through the use of intelligence, personality, and abilities.

Skill : Communication; liasion between management, users, and developers; assigning activities; monitoring progress.

b) **Management** : Getting projects completed through the effective utilization of resources.

Skill : Defining and sequencing activities; communicating expectations; assigning resources to activities; monitoring outcomes.

c) **Customer relations** : Working closely with customers to ensure that project deliverables meet expectations.

Skill : Interpreting system requests and specifications; site preparation and user training; contact point for customers.

Date

d) Technical problem solving : Designing and sequencing activities to attain project goals.
Skill : Interpreting system requests and specifications; defining activities and their sequence; making trade-offs between alternative solutions; designing solutions to problems.

e) Conflict management : Managing conflict within a project team to assure that conflict is not too high or too low.
Skill : Problem solving; smoothing out personality differences; compromising; goal setting.

f) Team management : Managing the project team for effective team performance.
Skill : Communication within and between teams; peer evaluations; conflict resolution; team building; self-management.

g) Risk and change management : Identifying, assessing and managing the risks and day-to-day changes that occur during a project.
Skill : Environmental scanning; risk and oppurtunity identification and assessment; forecasting; resource redeployment.

Date

Phases of Project Management

Project management is a controlled process of initiating, planning, executing, monitoring, controlling and closing down a project.

1. Initiating: During this phase, the project is conceptualized and feasibility is determined. Defining the project goal; defining the project scope; identifying the project manager and the key stakeholders; identifying potential risks; and producing an estimated budget and timeline are some activities performed on this phase.

2. Planning: In this phase, the project manager will create a blueprint to guide the entire project from ideation through completion. This blueprint will map out the project's scope; resources required; estimated time and financial commitments; communication strategy; execution plan etc.

3. Executing: During this phase, the project manager will conduct the procurement required for the project as well as staff the team. Execution of the project objectives are carried out to keep the entire project on time and on budget.

4. Monitoring and control: During this phase, project managers will closely measure the progress of the project to ensure it is developing properly. Documentation such as data collection and verbal and written status reports may be used. Monitoring will detect any necessary corrective action or change in the project to keep the project on track.

5. Closing : The closing process group occurs once the project deliverables have been produced and the stakeholders validate and approve them. During this phase, the project manager will close contracts with suppliers, external vendors, consultants and other third-party providers. All documentation will be archived and a final project report will be produced.

Representing and Scheduling Project Plans :

A project manager has a wide variety of techniques available for depicting and documenting project plans. These planning documents can take the form of graphical or textual reports, although graphical reports have become most popular. The most commonly used methods are Gantt charts and Network diagrams.

a) Gantt chart : Gantt chart is a graphical representation of a project that shows task activity as a horizontal bar which is proportional to its time for completion. Gantt charts are often useful for simple projects or sub-parts of a larger projects. It is also useful for monitoring the progress of activities. Gantt charts do not show how tasks must be ordered but simply show it should end.

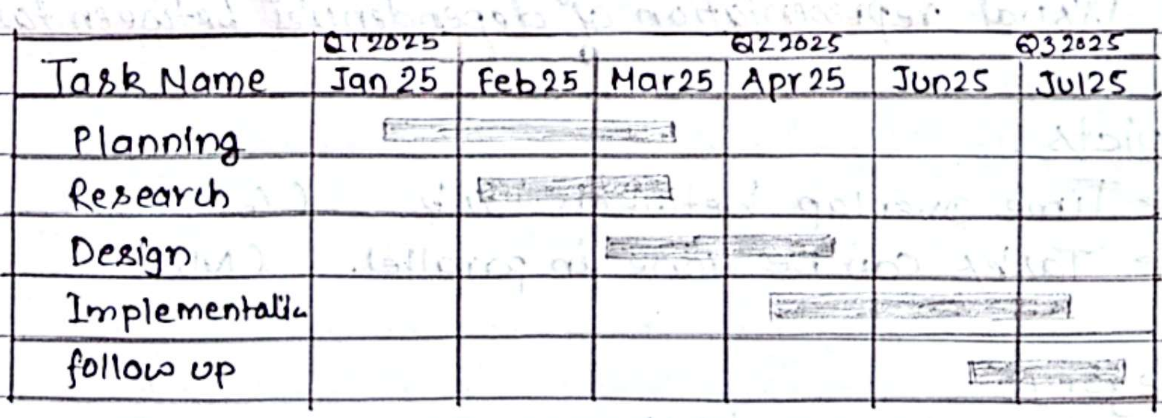


fig :- Typical example of Gantt chart.

Date

b) <u>Network Diagram:</u> Network diagram is a graphical representation of a project that shows how each activity relates to others in the project. The project manager can track each element of project and report progress to stakeholders with the help of network diagrams.
Network diagram shows the ordering of activities by connecting a task to its predecessor and successor tasks.

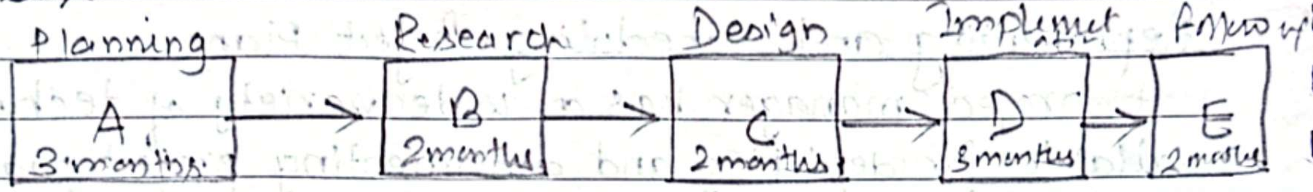


fig: Typical example of Network diagram.

<u>Comparison between Gantt chart and Network diagram.</u>

* <u>Definition:</u>
→ Gantt chart is a graphical representation that shows each task activity with their corresponding start and finish date.
→ Network Diagram is a schematic display of logical relationships among project activities.

** <u>Tasks:</u>
→ Visual representation of duration of tasks. (GC)
→ Visual representation of dependencies between tasks. (ND)

** <u>Depicts:</u>
→ Time overlap between task. (GC)
→ Tasks can be done in parallel. (ND)

* <u>Category:</u>
→ Bar chart (GC)
→ Flow chart (ND).

Date

* slack time:
  → Visually shows slack time. (GC)
  → Shows slack time by using rectangles filled by data.

Representing Project Plans

Project scheduling and management requires time, costs and resources to be controlled. Resources are any person, group of people, or material used in accomplising an activity. Network diagramming is a critical path scheduling technique used for controlling resources. A major strength of Network diagramming is its ability to represent how completion times vary for activities. Because of this it is more often used than Gantt charts to manage projects such as information system development where variability in the duration of activities is the norm.

Calculating Expected time durations using PERT:

PERT (Program evaluation review technique) is a technique that uses optimistic, pessimistic, and realistic time estimates to calculate the expected time for a particular task. This technique helps us to obtain a better time estimate when we are uncertain as to how much time a task will require to be completed.

$$ET = \frac{O + 4r + P}{6}$$

where,

ET = expected time completion for activity.
O = optimistic time completion for an activity.
P = pessimistic time completion for an activity.
r = realistic time completion for an activity.

Date

Using Project Management Software

A wide variety of automated project management tools are available to help to manage a development project. New versions of these tools are continously being developed and released by software vendors. Most of the available tools have a set of common features ~~resources~~ that include the ability to define and order tasks, assign resources to tasks, and easily modify tasks and resources. Project management tools are available to wide variety of systems. Most programs offer features such as PERT/CPM, Gantt charts, resource Scheduling and cost tracking. Asana, Wrike, Trello, Jira, clickUp, Microsoft Project etc are examples of project management software.

* Forward, reverse and round trip Engineering

These are the features provided by the CASE tools.

Generating or transforming system models into program code is forward engineering. System analyst draws the models from scratch or template and the CASE tools generates the program code.

Reverse Engineering allows a case tool to read existing program code and transform that code into a representative system model that can be edited and refined by the systems analyst.

CASE tools that allow for bi-directional forward and reverse engineering are said to provide for "round-trip engineering".

Date

Importance of Information System.

→ To capture and manage data to produce useful information that supports an organization and its employees.
→ To gain maximum benefits from company's information System.
→ To increase effictiveness and accuracy.
→ To make better decisions based on data present in the system.
→ For keeping records of a company for future use.